

# DDM512 Database Design & Implementation

## Practical Observation Checklist (Task 3)

Name of student  
being observed

Deacon George \_\_\_\_\_

Date 11<sup>th</sup> June, 2026 \_\_\_\_\_

| Task | Behaviour Observed                                                                                                                                                       | Tick | Comments On What Was Observed<br>(e.g., Confident, Hesitant, Confused)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1  | 1. Student creates a database to the design created in Task 2.                                                                                                           | ✓    | You successfully created the database according to your design and implemented all required tables with correct entities, attributes, data types, and constraints, including primary and foreign keys. You inserted at least three records per table and demonstrated encryption and decryption of sensitive data effectively. All records were displayed in plain text, and you correctly performed both single and multiple updates and deletions. Your submission includes two clear examples of each SQL command and at least two examples of data retrieval. Well done. |
|      | 2. Student creates all tables with their entities, attributes, data types, primary and foreign key constraints, and other relevant queries/constraints.                  | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 3. Student inserts at least THREE records for each table.                                                                                                                | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 4. Student encrypts and decrypts at least one piece of sensitive data.                                                                                                   | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 5. All records must be displayed in plain text.                                                                                                                          | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 6. Student updates and deletes single and multiple records of any table.                                                                                                 | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 7. Student provides TWO examples of each SQL command.                                                                                                                    | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 8. Student provides at least TWO examples of data retrieval                                                                                                              | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.2  | 9. Student creates TWO user roles and assigns each with specific access rights.                                                                                          | ✓    | Good work! You successfully created two user roles with appropriate access rights and added two MySQL users for each role. You applied at least two string functions for data cleansing, and correctly used SQL commands to check all tables, analyse a specific table, and optimize all tables in the database.                                                                                                                                                                                                                                                             |
|      | 10. Student creates TWO new MYSQL users for each user role with specified privileges.                                                                                    | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 11. Student cleanses the database using at least TWO string functions.                                                                                                   | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 12. Student selects and creates the SQL command to check all tables in the database.                                                                                     | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 13. Student selects and creates the SQL command to analyse a table in the database.                                                                                      | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 14. Student selects and creates the SQL command to optimise all tables in the database.                                                                                  | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.3  | 15. Student prepares at least four test cases to the listed specifications.                                                                                              | ✓    | You have prepared at least four test cases as required and clearly documented each key step. Well done.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|      | 16. Student documents each key step.                                                                                                                                     | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.4  | 17. Student performs an automated daily backup of the database and captures each step involved.                                                                          | ✓    | You successfully demonstrated the full backup and restore process by automating daily backups, securely storing them in line with organisational and industry standards, and restoring the database using the backup—all steps were clearly documented.                                                                                                                                                                                                                                                                                                                      |
|      | 18. Student securely stores backup(s) using security mechanisms that meet organisational requirements and industry best practices and captures each step of the process. | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | 19. Student restores the database using the backup and captures each step of the process.                                                                                | ✓    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Student's Name Deacon George \_\_\_\_\_ Signature *Deacon George* \_\_\_\_\_

Assessor's Name Tracy Fei \_\_\_\_\_ Signature Tracy Fei \_\_\_\_\_

